Anbang Wang

+86 18853138398 | awangas@connect.ust.hk

Wang Anbang | Aawangas

Hong Kong SAR, China

Research Interest

My research interests primarily focus on computer vision and machine learning. Specifically, I am particularly interested in developing fundamental methods to fully leverage large-scale world knowledge. My recent interests are particularly centered on 3D vision.

Education

- **The Hong Kong University of Science and Technology** *2023/9 - 2027/9 (expected)*
  BSc in data science and technology Hong Kong SAR, China
  - GPA: 4.035/4.3 (top 2%)
- **École Polytechnique Fédérale de Lausanne** *2026/2 - 2026/7 (expected)*
  Exchange Program in Computer Science section Lausanne, Switzerland

Publications

- **SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation** *ECCV 2026*
  Anbang Wang*, Yuzhuo Ao*, Shangzhe Wu, Chi-Keung Tang
  ** Equal contribution.*
- **ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation** *arXiv*
  Yuzhuo Ao*, **Anbang Wang***, Yu-Wing Tai, Chi-Keung Tang
  ** Equal contribution.*
- **MedSapiens: Taking a Pose to Rethink Medical Imaging Landmark Detection** *MedIA 2026*
  Marawan Elbatel*, **Anbang Wang***, Keyuan Liu, Kaouther Mouheb, Enrique Almar-Munoz, Lizhuo Lin, Yanqi Yang, Karim Lekadir, Xiaomeng Li
  ** Equal contribution.*
- **Geometric-Guided Few-Shot Dental Landmark Detection with Human-Centric Foundation Model** *MICCAI 2025*
  Anbang Wang*, Marawan Elbatel*, Keyuan Liu, Lizhuo Lin, Meng Lan, Yanqi Yang, Xiaomeng Li
  ** Equal contribution.*

Projects

- **SK-Adapter: Skeleton-Based Structural Control for Native 3D Generation** *2025/11 - 2026/03*
  Advised by Prof. Chi-Keung Tang and Prof. Shangzhe Wu []
  - Developed SK-Adapter, a lightweight skeleton-based adapter that injects joint-coordinate and topology tokens into a frozen native 3D generation backbone via cross-attention.
  - Introduced skeletons as first-class structural control signals for precise 3D asset generation, reducing ambiguity from text- or image-only prompts.
  - Contributed to Objaverse-TMS, a 24k text-mesh-skeleton dataset, and demonstrated robust skeleton-conditioned generation and local 3D editing while preserving geometry and texture quality.
- **ReasonNavi: Human-Inspired Global Map Reasoning for Zero-Shot Embodied Navigation** *2025/06 - 2025/09*
  Advised by Prof. Chi-Keung Tang And Prof. Yu-Wing Tai []
  - Developed ReasonNavi, a framework that leverages MLLMs for global reasoning and a deterministic planner for local navigation, creating a novel reason-then-act paradigm.
  - Enabled unified, zero-shot navigation across diverse tasks (object, image, text goals), eliminating the need for task-specific fine-tuning or reinforcement learning.
  - Demonstrated superior performance over existing methods in efficiency and reliability.
- **Geometric-Guided Few-Shot Dental Landmark Detection with Human-Centric Foundation Model** *2024/12 - 2025/03*
  Advised by Prof. Xiaomeng Li []
  - Engineered a novel few-shot learning framework (GeoSapiens) by creatively adapting a human-centric foundation model (Sapiens) for a challenging medical imaging task.
  - Innovated a novel geometric loss function that encodes anatomical structural priors, significantly improving the model's robustness and precision in low-data scenarios.
  - Achieved state-of-the-art (SOTA) results, outperforming the previous leading method by 8.18% in Success Detection Rate (SDR) at the clinically-critical 0.5mm precision threshold.
- **MedSapiens: Taking a Pose to Rethink Medical Imaging Landmark Detection** *2024/11 - 2025/11*
  Advised by Prof. Xiaomeng Li []

- Developed MedSapiens by adapting a large-scale human-centric foundation model (Sapiens) for medical anatomical landmark detection, leveraging its inherent spatial pose localization priors across diverse imaging modalities.
- Harmonized heterogeneous datasets across four distinct anatomical regions (head, hand, chest, and legs) into a unified training pipeline, utilizing LoRA and heatmap-based decoding to capture shared spatial hierarchies across diverse clinical tasks.
- Established new state-of-the-art (SOTA) performance, outperforming generalist models by 5.26% and specialist models by 21.81% in Success Detection Rate (SDR), while achieving a 2.69% improvement in few-shot generalization on novel tasks.

- **Unsupervised Domain Adaptation for 3D Bone Reconstruction from 2D X-rays** *2024/09 - 2024/11*
  Advised by Prof. Xiaomeng Li
  - **Investigated** the challenge of domain shift in 3D bone reconstruction by building a baseline model utilizing the Pixel-Aligned Implicit Function (PiFU) framework.
  - **Implemented** a state-of-the-art unsupervised domain adaptation (UDA) algorithm to enhance the model's generalization capabilities across different X-ray datasets.
  - **Aimed to improve** reconstruction accuracy and robustness on out-of-domain data without requiring costly new annotations from the target domain.
- **Knowledge Distillation on Skin Lesion Segmentation** *2024/01 - 2024/09*
  Advised by Prof. Albert C. S. Chung [ ]
  - **Developed** a lightweight segmentation model for skin lesions by designing and implementing a knowledge distillation (KD) pipeline.
  - **Engineered** a student-teacher framework where knowledge from a large, complex model was effectively transferred to a compact student network.

Honors and Awards

- **HKUST Scholarship for Continuing Undergraduate Students (highest band)**
- **UROP research travel sponsorship**
- **Dean's List for all semesters**

Teaching Experience

- **Teaching Assistant of COMP 2711, Discrete Math and its Applications in Computer Science** *2024/09 - 2024/12*
  HKUST, CSE department
  - **Assist** weekly tutorial sessions for undergraduate students, clarifying core concepts in logic, set theory, graph theory, and combinatorics.
  - **Provided** regular one-on-one guidance, helping students debug complex proofs and master challenging assignment problems.

Academic Engagement and Service

- **Reviewer**, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- **Reviewer**, European Conference on Computer Vision (ECCV), 2026
- **Attendee**, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Daejeon, Republic of Korea, Oct 2025

Skills

- **Programming Languages:** Python (Proficient), C/C++ (Proficient)
- **Frameworks & Libraries:** PyTorch, NumPy, Pandas, Scikit-learn, OpenCV
- **Developer Tools & OS:** Git, GitHub, Linux, LaTeX
- **Languages:** Mandarin (Native), English (Fluent - IELTS 7.5/9.0)